9.1 INTRODUCTION

The science of man is also changing and human physiology has changed a lot since the time, the branch of pharmacology began. Science in general changed completely as far back as 1925 with Werner Heisenberg's uncertainty principle. Drug administration is the major responsibility of the nurses. To handle the drugs, nurses should have the basic knowledge of action, adverse effects, indication and contraindication of commonly used drugs and nursing responsibilities while administering the drug.



9.2 DEFINITIONS

Pharmacology is the study of the interaction of chemicals with living systems. The term pharmacology is obtained from the Greek word "pharmakon" which means as drug and "logos" means the study of science.

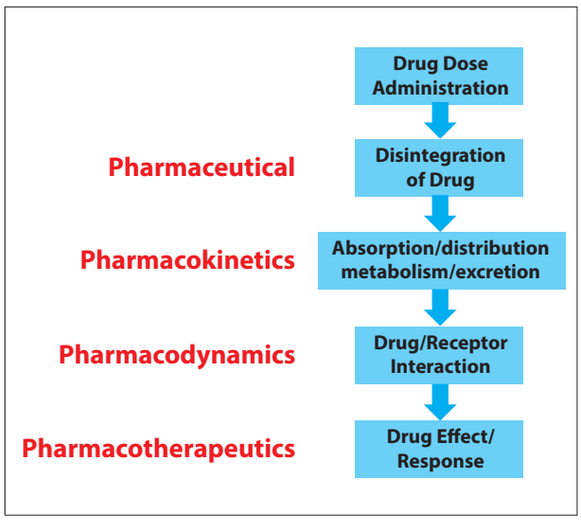
Drugs are chemicals that act on living systems at the chemical (molecular) level. The term drug is derived from the French word "drogue" which means "dry herb".

Medical pharmacology is the study of drugs used for the diagnosis, prevention, and treatment of disease.

Toxicology is the study of the untoward effects of chemical agents on living systems. It is usually considered an area of pharmacology.

Pharmacodynamic properties of a drug describe the action of the drug on the body, including receptor interactions, dose-response phenomena, and mechanisms of therapeutic and toxic action.

Pharmacokinetic properties describe the action of the body on the drug, including absorption, distribution, metabolism, and excretion. Elimination of a drug may be achieved by metabolism or by excretion



9.3 BASIC KNOWLEDGE REGARDING DRUG

The Nature of a Drug

The drug molecule interacts as an agonist (activator) or antagonist (inhibitor) with a specific molecule in the biologic system that plays a regulatory role. This molecule is called a receptor. Drugs may be synthesized within the body like hormones or may be chemicals not synthesized in the body like xenobiotics.

Poisons are also drugs that have exclusively harmful effects. Toxins are usually defined as poisons of biologic origin, i.e., synthesized by plants or animals. A drug is

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